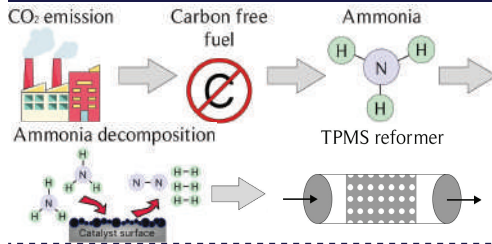


## Motivation



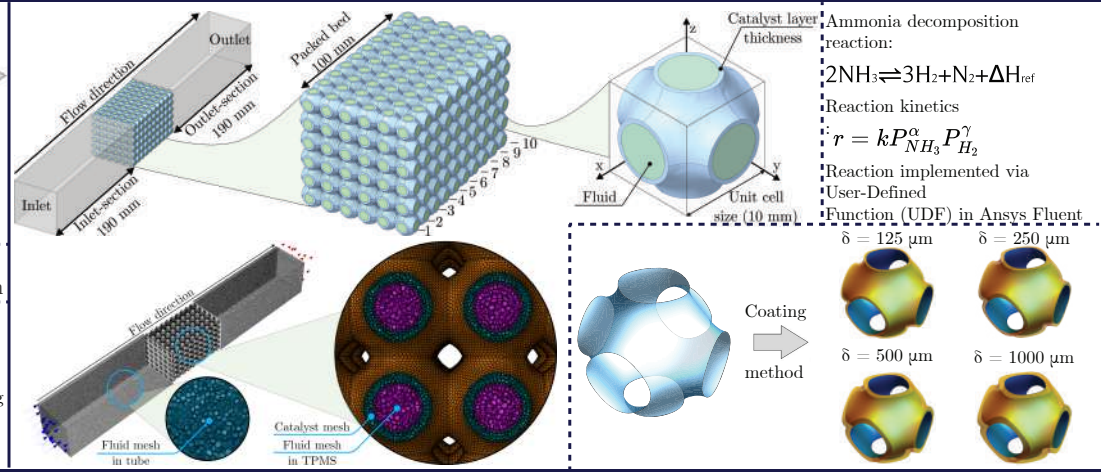
Motivation: Develop compact ammonia efficient reformer with TPMS-based structure for hydrogen-rich gas generation

Triply Periodic Minimal Surfaces (TPMS) advantages:

1. High specific heat transfer surface
2. Lightweight
3. Efficient material distribution
4. Controlled porosity and other properties

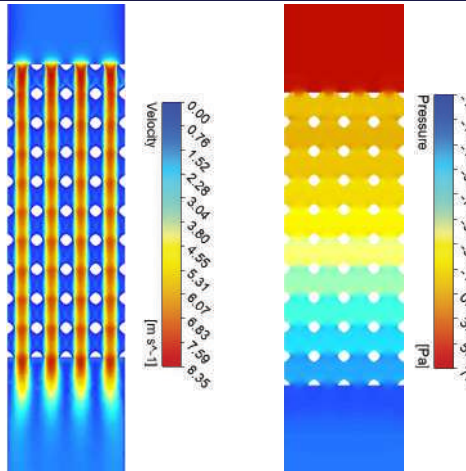
Ammonia's low laminar burning velocity (7 cm/s) limits its fuel potential. The solution is on-demand hydrogen production, yielding a much higher velocity (207 cm/s) comparable to gasoline (35 cm/s).

## Methods

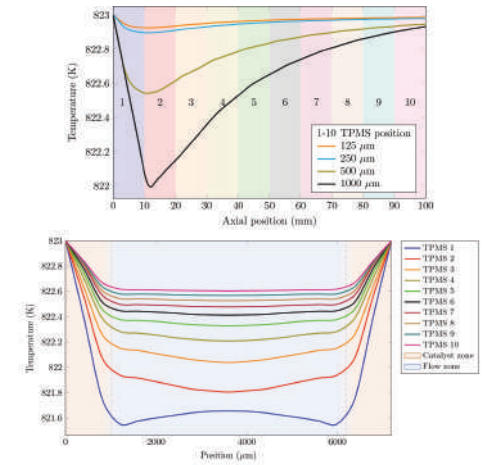
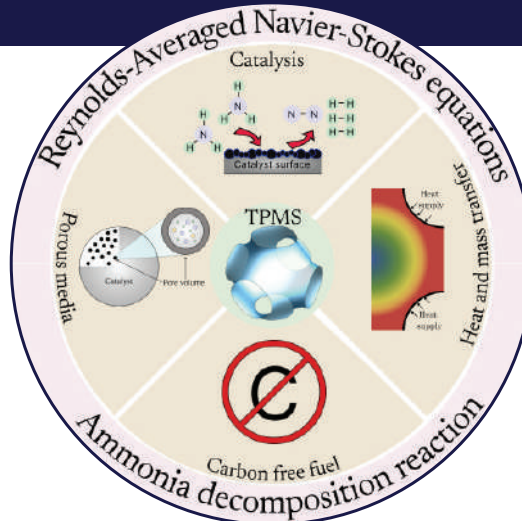


## Results

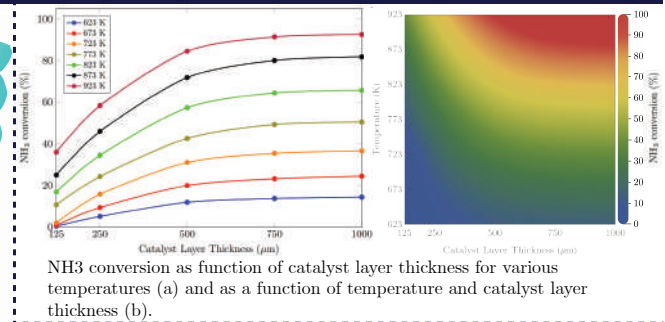
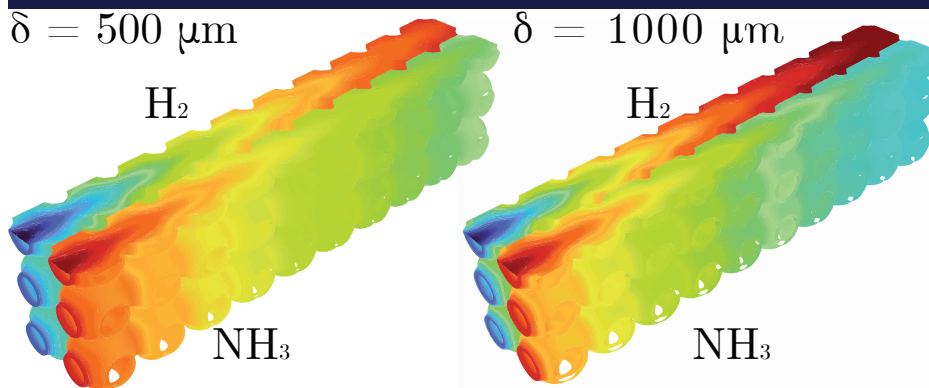
### 1. Flow dynamics



### 2. Heat transfer



### 3. Chemical reaction



The NH<sub>3</sub> and H<sub>2</sub> molar fractions contours at T = 823 K for various catalyst layer thicknesses: 500 μm and 1000 μm.

At the temperature of 623 K, the catalyst layer thickness shows a minimal influence on the ammonia decomposition rate, primarily due to the reaction is kinetically limited. With a catalyst layer of 125 μm at 623 K, the conversion is negligible (near zero), while increasing the thickness to 1000 μm yields a conversion of only approximately 11%.

## Conclusion and future work

1. The impact of catalyst layer thickness is heavily dependent on temperature; it is negligible at lower temperatures (623–673 K) due to kinetic limitations, but becomes a primary driver of near-complete decomposition at higher temperatures (>923 K).
2. For the specific 10 mm Schwarz-P unit cell tested, 500 μm is the optimal catalyst layer thickness, effectively balancing maximum active site availability with efficient material utilization.
3. The model indicates that increasing the catalyst thickness beyond 500 μm (up to 1000 μm) yields diminishing returns, as internal mass transfer resistance prevents reactants from reaching the deeper catalyst layers.

Using others TPMS structures with a Ni-Al<sub>2</sub>O<sub>3</sub> catalyst for ammonia decomposition process

